Hard Coral Cover Model

Equation 1

$$C_t^i = \min\Big(K^i,\ \max\big(0,\ (C_{t-1}^i + G_t^i + S_t^i)(1-m^i)\big)\Big)$$

with the growth and recruitment terms

$$G_t^i = \max\big(0,\ r^i(1-g^i)\big)\, C_{t-1}^i \Big(1 - \frac{C_{t-1}^i}{K^\tau}\Big),$$

$$S_t^i = \frac{100\,\mu\, c}{A^i}\Big(\sum_j p_{j\to i}\,\Lambda_{t-1}^j\Big)\Big(1 - \frac{C_{t-1}^i}{K^\tau}\Big),$$

the larval supply

$$\Lambda_{t-1}^j = L\, C_{t-1}^j + R\cdot \mathbf{1}\{\, t=1,\ j\in\mathcal{T}\,\},$$

and the mortality fraction

$$m^i = \min\Big(0.95,\ P^{i,\mathrm{Typh}}\mathrm{mort}^{i,\mathrm{Typh}} + P^{i,\mathrm{Ble}}\mathrm{mort}^{i,\mathrm{Ble}}\Big).$$

Term Definitions

Symbol	Code variable	Definition
C_t^i	`current_hcc`	Hard coral cover of reef i at step t
C_{t-1}^i	`current_hcc`	HCC of reef i in the previous step
C_{t-1}^j	—	HCC of source reef j in the previous step
G_t^i	`growth`	Logistic growth term for reef i at step t
S_t^i	`recruit`	Recruitment (settled larvae) term for reef i
r^i	`R_GROW`	Intrinsic growth rate (by reef type)
g^i	`PENALTY/p_vec`	Gravity penalty on growth (code only), $0 \le g^i \le 1$
K^i, K^τ	`K_VALS`	Carrying capacity / max HCC for reef type τ
m^i	`EXP_MORT`	Expected mortality fraction (rate on standing cover)
$M^{i,\mathrm{Typh}}$	—	Expected HCC lost to typhoons $= P^{Typh}\mathrm{mort}^{Typh}C_{t-1}^i$
$M^{i,\mathrm{Ble}}$	—	Expected HCC lost to bleaching $= P^{Ble}\mathrm{mort}^{Ble}C_{t-1}^i$
$P^{i,\mathrm{Typh}}$	`typhprob`	Typhoon probability at reef i
$P^{i,\mathrm{Ble}}$	`bleachprob`	Bleaching probability at reef i
$\mathrm{mort}^{i,\mathrm{Typh}}$	`typhmort`	Per-event typhoon mortality fraction
$\mathrm{mort}^{i,\mathrm{Ble}}$	`bleachmort`	Per-event bleaching mortality fraction
$p_{j\to i}$	`P_T` entry	Settlement probability, source j to sink i
$L^{\,j\to i}$	—	Larvae received by i from j (photo term)
Λ_{t-1}^j	`larvae`	Larval supply from source j (code term)
A^i	`AREA`	Area of reef i (km^2)
L	`LARVAE_PER_HCC`	Larvae per unit HCC $= 10^6$
μ	`MATURITY_RATE`	Maturity/settlement coefficient $= 0.09$
c	`COLONY_TO_KMSQ`	Colony-to-area conversion $= 2\times 10^{-10}$
R	`RESTORATION_LARVAE`	Restoration pulse $= 1.4\times 10^9$ (at $t=1$, targets)
$\mathcal{T}$	`target_indices`	Set of targeted reefs
$\mathbf{1}\{\cdot\}$	—	Indicator: 1 if condition true, else 0